

The glide axioms & transforms — one congruence, two roles, four laws

The whole family, put in order: one primitive — a line congruence with a role (edges or rays) — four axioms for the edge role, one transform (the inversion T that swaps the role), and everything else demoted to theorem. Two frames on the drawing — a picture frame that draws true-size and a depth frame that is its translated, stretched copy — one shared station, and a projective measure. The three pins use up all three Möbius freedoms, so the law $\lambda = k/(k+1)$ is not chosen: it is forced. Drag anywhere — the two frames, the aperture plate, and the light test move as one object. The companion paper's single dial D is a path across this square ($r_u = 1-k$, $r_v = 1$), not a second axis.

presets: general glide · isotropic · vanishing-axis · one-point · pushbroom · orthographic

Left = the two frames (drag a purple corner) · Middle = the aperture plate (drag the marker) · Right = the light test (drag to orbit) · arrow keys nudge — interactive canvas on the live page; not embedded in this document

The axioms

	Axiom	What it pins
0	The primitive. A line congruence from the catalog — the parallel bundle, a pencil, or a hyperbolic linear congruence (two skew focal lines: the directing axes) — together with a role: edges or rays.	What every reading shares; the (r_u, r_v) square charts the catalog. $P \cdot D \cdot S \cdot M$ below are the laws of the edge role.
P	The $z = 0$ copy is the identity: the picture plane draws true-size. A square is a square.	$\lambda(0) = 0$ — depth zero sits on the picture frame.
D	Every depth draws as an axis-aligned, translated, stretched copy of the picture grid — grid parallel to grid at all depths. The far copy is the depth frame.	Supplies the two channels; $\lambda(\infty) = 1$ — the far world sits on the purple frame.
S	One station. A single depth, shared by both channels, maps beyond the frame.	$\lambda(\text{station}) = \infty$ — the third and last Möbius freedom.
M	Each channel reads depth projectively as a whole — scale and offset together: every channel coordinate is a Möbius function of depth.	λ is a Möbius map of the whole channel map; the pins then force everything.

The theorems

Theorem	Verified in
T1 · The law. $\lambda = k/(k+1)$ on every line, at every rate — no freedom remains after the pins.	check_axioms 1
T2 · Straight tracks. The shared station keeps every depth-track a straight line in the picture.	check_axioms 2
T3 · Cameras carry no frame. Split the station and the far image collapses to a point — the cross-slit camera has no depth frame.	check_crossslit_frame 1-2
T4 · Two frames are the whole datum. Picture frame + depth frame force every mark of the projection.	check_two_planes 2

Theorem	Verified in
T5 · The directing fates. Merge into the limit ($r = 1$) · survive finite ($0 < r < 1$) · leave for infinity ($r = 0$). The surviving directing point is the bend.	check_axioms 3
T6 · The light theorem. Fibers are straight at the four corners only — pinhole, orthographic, the two pushbrooms. The interior is impenetrable.	check_ray_bend 3
T7 · The armature identity. The glide lines are the rays of the cross-slit congruence of the two directing axes: the frame is cross-slit, the law is the glide.	check_ray_bend 2
T8 · The inversion is an involution. T swaps the role on the same congruence, and $T^2 = \text{id}$: each reading determines the congruence, the congruence determines the other reading, and the round trip glide $\rightarrow$ camera $\rightarrow$ glide is the identity.	check_transposition 1-2
T9 · The two fixed loci, strictly nested. Drawn lines are T-fixed exactly on the degenerate catalog — the diagonal (pencils), the $r = 0$ edges (slit + ∞), the corners; full maps only at the corners. $\text{Fix}_{\text{maps}} \subsetneq \text{Fix}_{\text{lines}}$	check_transposition 3-5

The directing fates

Rate	Name	The directing point	That channel's ray
$r = 1$	perspective channel	merges into the depth limit	straight
$0 < r < 1$	glide channel	survives — finite and separate	bent — the directing point is the bend
$r = 0$	orthographic channel	leaves for infinity	straight

Light passes $\Leftrightarrow$ no survivor on the line. Both channels must give up their directing points: exactly the four corners.

Transformational logic — the moves

Move	On the square	On the drawing
stretch the depth frame	slide the marker	the whole projection follows — the law is forced (T1, T4)
merge a directing point ($r \rightarrow 1$)	reach an $r = 1$ edge	the frame collapses to a segment; that channel straightens
release a directing point ($r \rightarrow 0$)	reach an $r = 0$ edge	the frame weds the picture in that axis; that channel straightens
surrender both	land on a corner	light passes: pinhole · orthographic · the two pushbroom slits
swap the channels	reflect across the diagonal	the drawing turns a quarter-turn (check_axioms 6)
split the station ($S \rightarrow S'$)	leave the square	the tracks curve, one law becomes two, the frame collapses to a point: the cross-slit camera (T2, T3)
keep the frame, change the law	stay at the same point of the square	the armature identity: the same lines carry either graduation (T7)

Move	On the square	On the drawing
transpose the role (T)	stay at the same point of the square	the same congruence re-read, edges $\nrightarrow$ rays: the shadow armature is shared always; the camera's tracks stay on the drawn lines only over the degenerate catalog; the full maps agree only at the corners; $T^2 = \text{id}$ (T8, T9)

Outside anchors

This system	Established name	Where
the corner-and-edge camera catalog	General Linear Cameras (GLC)	Yu-McMillan, ECCV 2004
two frames, linear interpolation, rank degeneration	ray-bundle cross-sections $P' = (1-d)P + dI$	Adams-Levoy, General Linear Cameras with Finite Aperture
the two-axis congruence; the directing axes	hyperbolic linear congruence; its two focal lines	classical line geometry (Plücker; Pottmann-Wallner, Computational Line Geometry)
reading 2 on that congruence	the Crossed-Slits (X-Slits) projection; perpendicular axis-parallel slits = POX-Slits	Pajdla 2002; Zomet-Feldman-Peleg-Weinshall, IEEE PAMI 2003
cameras as line congruences	"What is a Camera?"	Ponce, CVPR 2009
the convergent-solid ancestor	relief perspective — a central perspective collineation (image plane + vanishing plane); reading 1's orthographic flatten is the departure	classical descriptive geometry
flattened solids under parallel viewing	the (generalized) bas-relief ambiguity	Belhumeur-Kriegman-Yuille, IJCV 1999

Anchors are reconnaissance-level (a handful of searches), to be confirmed in a full literature pass. The transposition itself — one congruence, two roles, two nested fixed loci — was not found under any name.

Axiom S is the personality of the family. Keep the one station and you glide: the law is single, the tracks are straight, the depth frame survives, and the surface is closed to light everywhere but the corners. Split the station and you photograph: the law divides, the tracks curve, the frame collapses to a point, and a straight ray threads two slits. The square is the space of depth frames; the perimeter is where they degenerate; the corners are where light gets in. And over it all sits the inversion: the same congruence worn as edges or as rays — an involution whose drawn lines survive on the degenerate catalog and whose full maps agree only at the corners. The laws above are the edge role's half of that symmetry.

Verified in `check_axioms.js` (7: the pins force the law; the shared station gives straight tracks and splitting it curves them and destroys the frame; the directing trichotomy; light at the corners only; channel swap = the diagonal reflection; translation never bends a ray) and `check_axiom_gap.js` (5: stating D for the far plane alone admits a rotating counterexample — true-size picture, affine far frame, rotated in between — so D is strengthened to every depth, which supplies the channels and forces the glide scale uniquely) and `check_dof_audit.js` (4: the repair itself was still gappy — the offsets were unpinned, and a wandering-offset projection passed P, D and scale-M while curving its tracks; auditing all four depth-functions and requiring M of the whole channel map forces the two-plane interpolation uniquely) and `check_transposition.js` (6: $T^2 = \text{id}$ — the round trip glide $\rightarrow$ camera $\rightarrow$ glide is the identity; the congruence is T's invariant; drawn lines are T-fixed exactly on the degenerate catalog; full maps only at the corners; the nesting is strict; the catalog is

closed) — standing on check_inversion (10), check_two_planes (8), check_ray_bend (3), check_crossslit_frame (9), check_split (9), check_vaxis (8), check_isotropy (7) and check_atlas (8). The page re-runs six checks on every drag. Audit total 317 · snapshot 2026-07 · matches atlas page.

Rebuilt from The-glide-axioms-v4.pages (text and table cells recovered from the archive). Theorem numbering T1-T9 and the theorem↔check and anchor-row pairings are reconstructed from content cross-references (the moves table's T-citations); the live page's interactive canvas is not part of the source file.